Week 8 online class

Worksheet 5 Q12

Q12

Theorem 8.1. The Boundedness Theorem.

Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$. Then f is bounded.

Theorem 8.2. Extreme Value Theorem

Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$. Then f attains its bounds on $[a, b]$.

These two theorems will be applied in the proof of Rolle's theorem in Chapter 5.

Note that *Theorems 8.1 and 8.2* are <u>not true</u>

on open intervals, e.g.,

$f(x) = \frac{1}{x}$ is continuous on $\Big(0,1\Big)$ but unbounded.

Proof. We prove this by contradiction.

Suppose that f is not bounded.

This leads to two cases below,

either f is unbounded above or unbounded below.

Case 1. f is unbounded above

$$\Rightarrow \exists x_n \in [a,b] \ni f(x_n) > n \ \forall n \in \mathbb{N}$$

$$\Rightarrow \left(f\left(x_{n_k}\right)\right) \text{ is } \underline{\text{unbounded}}$$

But $\left(x_n\right)$ is bounded

$$\underset{\text{BWT}}{\Rightarrow} \exists \left(x_{n_k}\right) \ni \lim\left(x_{n_k}\right) = \alpha \in \left[a, b\right]$$

$$\underset{\text{Seq cont}}{\Rightarrow} \lim\left(f\left(x_{n_k}\right)\right) = f(\alpha)$$

$$\underset{\text{Definition}}{\Rightarrow} \left(f\left(x_{n_k}\right)\right) \text{ converges}$$

$$\underset{\text{Thm 5.4, Chap 2}}{\Rightarrow} \left(f\left(x_{n_k}\right)\right) \text{ is } \underline{\text{bounded}}.$$

Case 2. f is unbounded below

$$\Rightarrow \exists y_n \in [a,b] \ni f(y_n) < -n \ \forall n \in \mathbb{N} \Rightarrow \left(f\left(y_{n_k}\right)\right) \text{ is } \underline{\text{unbounded}}.$$

But (y_n) is bounded $\underset{\text{BWT}}{\Rightarrow} \exists \left(y_{n_k}\right) \ni \lim\left(y_{n_k}\right) = \beta \in [a,b]$

$$\underset{\text{Seq cont}}{\Rightarrow} \lim\left(f\left(y_{n_k}\right)\right) = f(\beta) \Rightarrow \left(f\left(y_{n_k}\right)\right) \text{ converges}$$

$$\underset{\text{Thm 5.4, Chap 2}}{\Rightarrow} \left(f\left(y_{n_k}\right)\right) \text{ is } \underline{\text{bounded}}, \text{ a contradiction.}$$

Theorem 8.2. Extreme Value Theorem

Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$.

Then f attains its bounds on $[a, b]$.

What do we need to prove?

To prove that there exists a point,
say d in $[a, b]$
such that $f(d)$ is the sup/max.

Also to prove that there exists a point,
say e in $[a, b]$
such that $f(e)$ is the inf/min.

Proof. The supremum and infimum of $S = \left\{ f(x) : x \in \left[a, b \right] \right\}$ exist because f is bounded by Theorem 8.1, $f(a), f(b) \in S$ and $S \subset \mathbb{R}$.

Case 1. We show that $\exists d \in [a,b] \ni f(d) = \alpha = \sup S$.

By Theorem 4.2 (i) of Chap 1,

$$\alpha = \sup S \Leftrightarrow \exists f(x) \in S \ni \alpha - \varepsilon < f(x) \leq \alpha \ \forall \varepsilon > 0$$

$$\Rightarrow \exists x_n \in [a,b] \ni \alpha - \tfrac{1}{n} < f(x_n) \leq \alpha \ \forall n \in \mathbb{N}$$

$$\underset{\text{Squeeze Thm}}{\Rightarrow} \lim(f(x_n)) = \alpha \underset{\text{Thm 8.3, Chap 2}}{\Rightarrow} \lim(f(x_{n_k})) = \alpha$$

For all $\varepsilon > 0$, there exists an $f(x) \in S$ such that $\alpha - \varepsilon < f(x)$.

When $\varepsilon = 1$, there exists an $f(x_1) \in S$ such that $\alpha - 1 < f(x_1)$.

When $\varepsilon = \frac{1}{2}$, there exists an $f(x_2) \in S$ such that $\alpha - \frac{1}{2} < f(x_2)$.

...

When $\varepsilon = \frac{1}{n}$, there exists an $f(x_n) \in S$ such that $\alpha - \frac{1}{n} < f(x_n)$.

...

$\Rightarrow \exists x_n \in [a,b] \ni \alpha - \frac{1}{n} < f(x_n) \leq \alpha \ \forall n \in \mathbb{N}$.

$$\left(x_n\right) \text{ is bounded}$$

$$\underset{\text{BWT}}{\Rightarrow} \exists \left(x_{n_k}\right) \ni \lim\left(x_{n_k}\right) = d \in \left[a,b\right]$$

$$\underset{\text{Seq cont}}{\Rightarrow} \lim\left(f\left(x_{n_k}\right)\right) = f(d)\,.$$

$$\Rightarrow f\left(d\right) = \lim\left(f\left(x_{n_k}\right)\right) = \alpha\ .$$

Case 2. We show that $\exists e \in [a,b] \ni f(e) = \beta = \inf\{f(x) : x \in [a,b]\}$. By Theorem 4.2 (ii) of Chap 1, $\beta = \inf\{f(x) : x \in [a,b]\} \Leftrightarrow \exists y_n \in [a,b] \ni \alpha + \frac{1}{n} > f(y_n) \geq \beta \ \forall n \in \mathbb{N}$.

$$\underset{\text{Squeeze Thm}}{\Rightarrow} \lim(f(y_n)) = \beta \underset{\text{Thm 8.3, Chap 2}}{\Rightarrow} \lim(f(y_{n_k})) = \beta .$$

(y_n) is bounded $\underset{\text{BWT}}{\Rightarrow} \exists (y_{n_k}) \ni \lim(y_{n_k}) = e \in [a,b] \underset{\text{Seq cont}}{\Rightarrow} \lim(f(y_{n_k})) = f(e)$.

Hence, $f(e) = \lim(f(y_{n_k})) = \beta$.